

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) Screenshot... 133747.png

Screenshot 2026-08-07 133747.png(image/png) - 142757 bytes, last modified: 8/7/2026 - 100% done
 Saving Screenshot 2026-08-07 133747.png to Screenshot 2026-08-07 133747 (3).png

◆ Gemini

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab

# Define points for perspective transformation
roi_points = np.array([(150, 200), (450, 200), (550, 500), (50, 500)])
target_points = np.array([(0, 0), (400, 0), (400, 600), (0, 600)])
M = cv2.getPerspectiveTransform(roi_points.astype(np.float32), target_points.astype(np.float32))

# --- Changes for Colab compatibility ---
# Instead of webcam (cv2.VideoCapture(0)), read an uploaded image.
# You had uploaded 'Screenshot 2026-08-07 133747 (3).png', using it as an example.
# Please change the filename if you want to use a different uploaded image.
img = cv2.imread("Screenshot 2026-08-07 133747 (3).png")

if img is None:
    print("Error: Could not load the image. Please check the filename and ensure it's uploaded")
else:
    # Resize image to match the video frame size for consistent transformation, if needed
    # Or, adjust target_points and M based on img.shape
    # For this example, let's assume original image size is compatible or transformation
    # If the image dimensions are significantly different from expected (e.g., 640x480 for video),
    # the perspective transform might not look as intended.
    rows, cols = img.shape[:2]

    # Apply perspective transformation to the static image
    dst = cv2.warpPerspective(img, M, (cols, rows)) # Use image dimensions for warpPerspective
```

```
# Display original and transformed images using cv2_imshow
cv2_imshow(img)
cv2_imshow(dst)

# Save the transformed image
cv2.imwrite('Perspective_Transformed_Image_Static.jpg', dst)
print("Perspective transformed image saved as 'Perspective_Transformed_Image_Static.j

# Removed cv2.waitKey(1), cap.release(), and cv2.destroyAllWindows()
# as they are not needed with cv2_imshow and for static image processing.
```


